

Supplementary Material for Physics-Informed State Reconstruction and Event-Source Inference From Sparse PMUs Under Limited Observability

JORGE LUIS MAYORGA TABORDA¹, YESSICA AFRICANO RODRIGUEZ¹, MASOUD BARATI²,
PAYMAN DEGHANIAN³, AND DAVID FELIPE CELEITA RODRÍGUEZ⁴

¹Universidad de los Andes, Bogotá, Colombia

²Department of Electrical and Computer Engineering, University of Pittsburgh, Pittsburgh, PA, USA

³Department of Electrical and Computer Engineering, The George Washington University, Washington, DC, USA

⁴Faculty of Engineering, Universidad de La Sabana, Chía, Colombia

ABSTRACT This supplement records the detailed literature matrix, electrical conventions, theoretical derivations, estimator equations, frozen experimental contracts, and secondary results supporting the main article. It does not add claims or change any test denominator. Every numerical table and empirical figure is generated from the same hashed evidence snapshot used by the main manuscript.

INDEX TERMS phasor measurement units, reproducibility, state estimation, event-source inference, supplementary material.

APPENDIX A DETAILED LITERATURE CAPABILITY MATRIX

Table 1 expands the family-level synthesis in the main paper. “Held source” means that the physical asset is absent from source-labeled training and tuning. The entries describe published tasks and assumptions; they are not a cross-paper performance ranking.

The direct competitors solve different contracts. State and unknown-input estimators define the latent reconstruction problem but not the event taxonomy. Learned localizers rank represented sources but rarely remove the physical asset from every source-labeled fitting operation. Model-based outage methods provide the closest likelihood precedent, while functional observability and placement studies provide the information tests. These distinctions determine the baselines and prevent cross-paper accuracy comparisons with incompatible targets.

APPENDIX B ELECTRICAL MODEL AND EVENT OPERATORS

Within a smooth network mode, the DAE is locally index one after fixing the reference angle. Differentiating the algebraic constraint gives

$$\dot{z} = -g_z^{-1}(g_x f + g_u \dot{u} + g_\theta \dot{\theta}). \quad (1)$$

This reduction is not applied across a topology switch or fault insertion, where algebraic variables may jump. Differential

states remain continuous unless a simulator-specific reset is declared.

With $Y_{\text{bus}} = G + jB$ and $V_i = |V_i|e^{j\vartheta_i}$, the network injections are

$$P_i = |V_i| \sum_j |V_j| (G_{ij} \cos \vartheta_{ij} + B_{ij} \sin \vartheta_{ij}), \quad (2)$$

$$Q_i = |V_i| \sum_j |V_j| (G_{ij} \sin \vartheta_{ij} - B_{ij} \cos \vartheta_{ij}). \quad (3)$$

For branch $e = (i, j)$ with complex tap t_e , series admittance y_e , and terminal shunts $y_e^{f,sh}$ and $y_e^{t,sh}$,

$$\begin{bmatrix} I_{e,f} \\ I_{e,t} \end{bmatrix} = \begin{bmatrix} (y_e + y_e^{f,sh})/|t_e|^2 & -y_e/t_e^* \\ -y_e/t_e & y_e + y_e^{t,sh} \end{bmatrix} \begin{bmatrix} V_i \\ V_j \end{bmatrix}. \quad (4)$$

The implemented PMU operator selects the actual terminal and orientation. Replacing this current with complete bus-injection current changes the likelihood.

Positive-sequence phasors define the balanced model. Negative- and zero-sequence components remain integrity or unbalance evidence. Ideal continuous-frequency channels satisfy

$$f_i = f_0 + \frac{1}{2\pi} \frac{d\angle V_i}{dt}, \quad \text{ROCOF}_i = \frac{1}{2\pi} \frac{d^2 \angle V_i}{dt^2}, \quad (5)$$

but an operational likelihood must reproduce the causal filtering of the PMU backend.

TABLE 1. Detailed capability comparison for the closest methodological works.

Work	State target	Event/source target	Physics mechanism	Sparse sensing	Held source	Integrity treatment
Katanic <i>et al.</i> [1]	Differential and algebraic DAE states	Unknown injectors	Incomplete nonlinear DAE and least squares	Yes	N/A	No
Nugroho <i>et al.</i> [2]	Differential and algebraic DAE states	Unknown input and input-sensor failure	Outlier-resistant DAE observer	PMU voltage/current	N/A	Failure tolerance
Qu <i>et al.</i> [3]	Machine state	Unknown field voltage and torque	Event-triggered nonlinear state/input estimator	Event-triggered PMUs	N/A	Covariance bounds
Ngo <i>et al.</i> [4]	Power-system states	None	Branch-current physics plus graph learning	Yes	N/A	No
Yue <i>et al.</i> [5]	Distribution-system state	Line outage	Grid-model-informed graph learning	Yes	Not stated	Outlier-resistant SE
Moshagh <i>et al.</i> [6]	Transmission-system state	Topology variation	Topology-aware graph learning	PMU-unobservable systems	Not a source task	PMU failure and bad data
Pandey <i>et al.</i> [7]	None	Detection, class, and source	Topology/equipment information	PMU network	Not stated	No
Yildiz and Abur [8]	None	Fault detection and location	OLS/CNN and PMU placement	Yes	Not stated	No
Li <i>et al.</i> [9]	None	Multi-family disturbance source	Topology/type-constrained STGCN loss	Yes	Not stated	No
Kummerow <i>et al.</i> [10]	None	Known/unknown event type and location	Siamese distance and learned rejection	Limited/full sets	Open-set rejection	Rejection boundary
Hu and Cheng [11]	None	Open-set event detection/classification	Generative window evidence and fusion	Streaming PMUs	Unknown-event category	No source posterior
Rimkus <i>et al.</i> [12]	Missing PMU trajectories	Downstream fault localization	Graph reconstruction before diagnosis	Yes	Imputer only	Corruption via imputation
Zhai <i>et al.</i> [13]	Full dynamical state	Line-attack location	Learned temporal dynamics	Yes	Across benchmarks	Attack target only
Main article	Hidden voltage field	All load supports with $K \leq 2$; historical eight-label ontology	DAE hypotheses and quadratic physical likelihood	Eight PMU buses	Physical bank has no source-label head	Physical/integrity variables separate

TABLE 2. Physical intervention operators represented by the common DAE formulation.

Family	DAE action at source ℓ	Candidate support	Severity coordinates	State continuity
Bus fault	Add a shunt primitive for the clearing interval	admissible buses	complex admittance or resistance/reactance	x continuous; z may jump
Line outage	Remove branch tap, series, charging, and shunt primitives	in-service branches	status and switching time	x continuous; z may jump
Generation change	Modify scheduled injection or a declared governor reference	generator buses	$\Delta P_G, \Delta Q_G$, duration	continuous input-driven response
Load change	Modify the active/reactive withdrawal model	load buses	$\Delta P_L, \Delta Q_L$, ZIP convention	continuous input-driven response

APPENDIX C METRICS, CAUSALITY, AND EVALUATION UNITS

For trajectory r , hidden-voltage relative error is

$$\text{HVRE}^{(r)} = \frac{1}{T_r |\mathcal{U}|} \sum_{k=1}^{T_r} \sum_{i \in \mathcal{U}} \frac{|\hat{V}_{i,k} - V_{i,k}|}{\max(|V_{i,k}|, \epsilon)}. \quad (6)$$

It is a trajectory-and-bus mean reported as a percentage; it is not the standard per-frame synchrophasor TVE conformity statistic. Detection, event-family macro-F1, exact physical Top-1/Top-3, cardinality, candidate-set coverage, and integrity-source Top-1 retain their own populations. A joint outcome is correct only when every applicable component is correct.

A causal estimate at k uses $y_{0:k}$. A fixed-lag smoother, complete-window score, or centered temporal feature is retrospective. For source generalization, the physical asset is absent from source-labeled training, feature fitting, threshold selection, and calibration. Intervals and significance tests use the complete trajectory, event parent, or rebuilt plant as the in-

dependent unit; samples within a 30-Hz stream are correlated observations, not replicates.

APPENDIX D PROOFS AND ADDITIONAL DIAGNOSABILITY RESULTS

A. FUNCTIONAL RECONSTRUCTION AND COVARIANCE

Two initial states produce the same output history exactly when their difference belongs to $\ker \mathcal{O}_N$. Their requested outputs agree for every such pair exactly when that difference also belongs to $\ker L_r$. This proves the functional condition in the main paper. Stacking $L_0, \dots, L_N$ gives the complete-sequence statement.

Let $G_N = \mathcal{O}_N^T R_N^{-1} \mathcal{O}_N$. When the functional condition holds, the minimum-variance linear unbiased target estimate and covariance are

$$\hat{\delta z}_r = L_r G_N^\dagger \mathcal{O}_N^T R_N^{-1} \delta Y_N, \quad (7)$$

$$\Sigma_{z,r}^* = L_r G_N^\dagger L_r^T. \quad (8)$$

The kernel inclusion implies $L_r = L_r G_N^\dagger G_N$, which proves unbiasedness. Any other unbiased linear map adds Q satisfy-

ing $Q\mathcal{O}_N = \mathbf{0}$ and therefore adds the positive-semidefinite covariance term $QR_NQ^\top$. For a nonlinear DAE, this local result requires a constant-rank neighborhood condition before it becomes a functional-observability certificate [14].

B. COMPOSITE CANDIDATES, INTEGRITY NUISANCE, AND PMU ADDITION

For candidate h , define the affine mean set

$$\mathcal{M}_h(S) = s_{h,H} + \text{col}[\mathcal{O}_{h,H}, \mathcal{E}_{S,H}]. \quad (9)$$

The profiled margin is zero exactly when the response difference belongs to the combined nuisance column space. At that point, the two candidates have identical means and covariance; their two conditional error probabilities sum to one. This proves the local indistinguishability proposition.

When the corruption support is unknown but at most q channels are affected, the conservative margin is

$$D_{ij}^{(q)}(H) = \min_{S: |S| \leq q} D_{ij,S}(H). \quad (10)$$

This union-of-subspaces form connects event localization to sparse attack identification and unknown-input reconstruction [15], [16]. For a fixed nuisance contract, adding a conditionally independent PMU block adds a nonnegative residual term for every nuisance vector. Minimizing after augmentation preserves the inequality

$$D_{ij,S}^{(\mathcal{P} \cup \{p\})}(H) \geq D_{ij,S}^{(\mathcal{P})}(H). \quad (11)$$

Equality with a sum of separately profiled margins need not hold because both blocks share nuisance variables.

C. SPARSE INJECTIVITY AND OBSERVATION TIME

If two K -sparse event vectors produce the same history, their difference is at most $2K$ sparse and lies in $\ker \mathcal{H}_T$. Conversely, any nonzero vector in $\ker \mathcal{H}_T \cap \mathcal{A}_{2K}$ can be written as the difference of two K -sparse vectors that produce the same output. This proves the sparse-injectivity proposition.

For competing supports S and R , define

$$\Delta_{S \rightarrow R}^2(T) = \min_b \|\mathcal{H}_{T,sa} - \mathcal{H}_{T,Rb}\|_{R_T^{-1}}^2. \quad (12)$$

Under prefix-consistent causal whitening and a fixed nuisance domain, appending samples adds a nonnegative residual term before minimization. The separation cannot decrease. It grows linearly when the optimally profiled steady signatures differ and can saturate when only a square-summable transient distinguishes them. Thus observation time improves finite information but cannot repair an exact shared response.

APPENDIX E ESTIMATOR AND LIKELIHOOD DETAILS

A. CANDIDATE-CONDITIONED JOINT OBJECTIVE

For candidate h , define the residual $r_{h,t} = M_t[y_t - h_{\mathcal{P}_t}(x_t, z_t) - b_t]$. The negative twice-log joint density, excluding the hypothesis prior, is

$$\begin{aligned} \mathcal{J}_h^\circ &= \|x_0 - \bar{x}_0\|_{P_0^{-1}}^2 + \sum_{t=0}^k \|r_{h,t}\|_{R_t^{-1}}^2 \\ &+ \sum_{t=0}^{k-1} \|x_{t+1} - F_h(x_t, z_t)\|_{Q^{-1}}^2 + 2\lambda_b \sum_{t=0}^k \|b_t\|_1, \end{aligned} \quad (13)$$

subject to the candidate DAE constraints. The normalized evidence integrates $\exp(-\mathcal{J}_h^\circ/2)$ over the latent trajectory and corruption coefficients. A Laplace approximation multiplies the optimized density by the pseudo-determinant on locally identifiable directions and by $p(h)$ exactly once. If the determinant or normalized density is omitted, the result is an evidence score rather than a posterior probability.

B. STATE HIERARCHY, CALIBRATION, AND RECENTERING

B0 returns the equilibrium. B1 applies the prior-conditioned snapshot gain

$$K_{\text{LMMSE}} = P_0 C^\top (C P_0 C^\top + R)^{-1}, \quad (14)$$

$$\hat{x}_k^{\text{B1}} = K_{\text{LMMSE}}(y_k - y_0). \quad (15)$$

B2 carries the state and covariance between frames using the Kalman recursion in the main paper. B3 applies the exact linear-Gaussian backward recursion at fixed lags from one frame to 2 s and is retrospective [17].

The nominal uncertainty correction uses

$$\Sigma_{i,k}^{\text{U2}} = T \Sigma_{i,k} T^\top, \quad T = \text{diag}(\sqrt{\tau_{\mathbb{R}}}, \sqrt{\tau_{\mathbb{I}}}), \quad (16)$$

with temperatures fitted on independent calibration trajectories. The adequacy statistic is a five-frame trailing mean of

$$a_k = (\nu_k - \mu_\nu)^\top \Sigma_\nu^{-1} (\nu_k - \mu_\nu). \quad (17)$$

Hidden truth labels whether the state error is unacceptable only after inference; it never enters the statistic.

For physical recentering, d collects active/reactive changes at admissible PQ loads and active-power changes at PV buses. Development data select $\gamma = 10^3$, $q_d = 1$, and $\lambda = 0.01$. If J_{yd} and J_{zd} are the observed and hidden voltage Jacobians after eliminating voltage through the AC constraints, the target residual

$$r_{d \rightarrow z} = \frac{\|J_{zd} N(J_{yd})\|_F}{\max(\|J_{zd}\|_F, \epsilon)} \quad (18)$$

tests whether PMU-uninformed nuisance directions alter the requested voltage. A zero value would be required for exact local identifiability.

C. LOAD LIKELIHOOD CONFIGURATION

The W2 covariance combines the frozen channel covariance with an AR(1) temporal model estimated from equilibrium records with synthetic measurement noise. The cardinality

prior is $p(K = 0, 1, 2) = (0.20, 0.50, 0.30)$, supports are uniform conditional on K , and active amplitudes are independent $\mathcal{N}(0, 0.05^2)$. Operational integration uses 31-node Gauss–Hermite quadrature per active amplitude. Two-dimensional Gauss–Kronrod calculations provide an independent numerical check. The configuration excludes ambient load trajectories, instrument/timing-error sweeps, PMU loss, topology changes, and unknown onset.

APPENDIX F COMPLETE FROZEN EXPERIMENTAL CONTRACTS

The PowerDynamics validation ladder checks static translation, PMU terminal-current orientation, equilibrium residual, local modes, nonlinear acceptance, and first-order response before estimator scoring. All 980 main mismatch cases and 175 refinement cases pass the declared acceptance gates. The physical-recentering TEST includes 20 equilibria at each of three shift scales; a solve is accepted only when the optimizer succeeds, outputs are finite, and the maximum AC residual is at most 5×10^{-5} . Rejections remain in the denominator.

For GLOBAL-137, the physical dictionary, covariance, priors, onset, and quadrature are frozen before new trajectories are generated. Three disjoint noise realizations accompany each event path, and 100 additional H_0 records represent no event. The identifiers have zero overlap with 1,816 historical records. The T120 bank preserves exact prefixes at 30, 60, 90, and 120 frames; independent operating-point validation remains pending.

The ANDES ordinary split uses three replicas for training, one for development, and one for testing at each source configuration. The source-holdout audit removes the physical asset from every source-labeled operation and supplies the true event family at evaluation. These contracts answer interpolation and source generalization separately.

TABLE 3. Complete frozen-contract inventory. The main paper groups these contracts by research question.

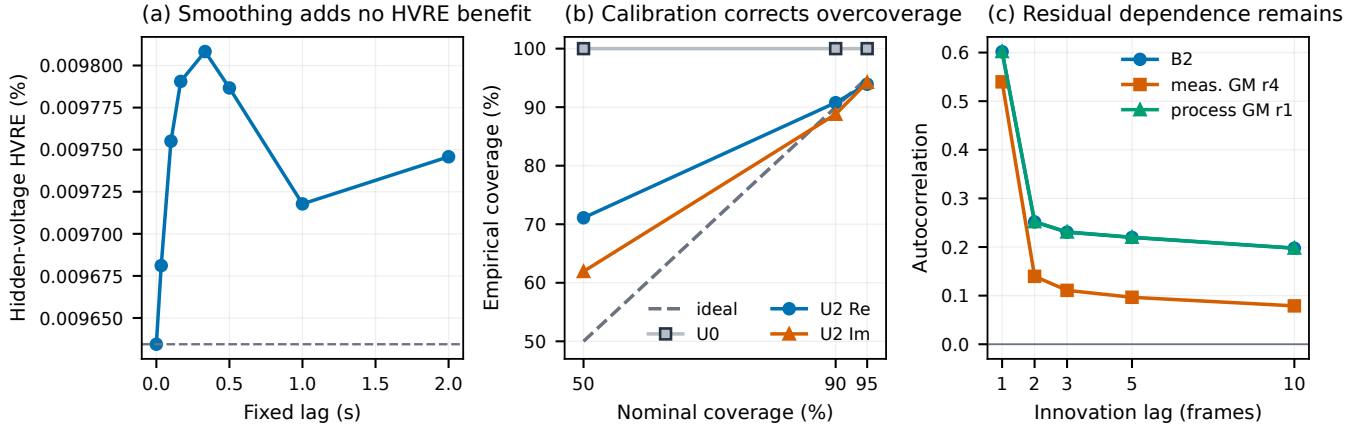
Study	Frozen contract	Independent evaluation unit
Structural audit	Preregistered voltage/current target residual, horizons 0–180	31 hidden buses
Nominal reconstruction	20 DEV / 100 TEST; 91 frames at 30 Hz; B0–B3 frozen	Complete nonlinear trajectory
Nominal calibration	30 independent CAL trajectories; U1/U2 and discrepancy only	Complete nonlinear trajectory
Physical mismatch	Seven families and seven scales; B1/B2/U2 frozen	980 rebuilt plants
Breakpoint refinement	New seeds in transition bands; B2 frozen	175 rebuilt plants
Causal recentering	M1/M2/M6/M7; 160 DEV / 320 TEST; tune on DEV	Rebuilt plant trajectory
Regularized physical recentering	M6 PQ/PV/Slack basis; 30 DEV / 60 TEST; one snapshot	Rebuilt equilibrium
Load tangent audit	16 load buses; centered $\pm 0.5\%$ nonlinear simulations	Candidate-bus intervention
Bayesian load inference V1	Eight amplitudes; 2,560 event / 320 no-event TEST records	Source-amplitude-noise record
Bayesian load inference V2	1,000 normal CAL; 500 normal + 640 event DEV; weak and finite TEST blocks	Source-amplitude-noise record
Prospective 137-hypothesis inference	Fresh H_0 + 16 singles + 120 pairs; 3,488 paths; 10,564 decisions	Physical trajectory or complete record, as declared
Weak-regime mechanism audit	Frozen GH31 prefixes at 5–30 frames; 2,400 paths	Retrospective source-pair trajectory
Long-horizon diagnostic	Exact prefixes through 120 frames; 32 nominal pair paths; 137 hypotheses	Diagnostic trajectory, not population recovery
Known-source event diagnosis	Eight abnormal labels plus normal; 414 TRAIN / 138 DEV / 138 TEST	Complete ANDES trajectory
Source-transfer audit	Asset-disjoint folds; true family supplied	101 event parents, each evaluated across three seeds

APPENDIX G SECONDARY RESULTS

The main paper retains the figures needed for the scientific argument. This appendix records secondary diagnostics and exact values that would otherwise interrupt that argument.

TABLE 4. Frozen TEST uncertainty and colored-discrepancy ablations. U2 is fitted on a disjoint 30-trajectory calibration set; Gauss–Markov (GM) ranks are selected on that calibration set.

Variant	Hidden NLL	95% cov. Re (%)	95% cov. Im (%)	Innovation ACF(1)	HVRE (%)
U0 original	-13.477	100.0	100.0	0.602	0.009634
U2 Re/Im temperature	-16.184	93.9	94.2	0.602	0.009634
D2-M measurement GM r4	14.991	66.6	48.1	0.540	0.009788
D2-P process GM r1	-13.475	100.0	100.0	0.602	0.009638

**FIGURE 1.** Secondary temporal and uncertainty diagnostics for B2/B3 and calibration variants. These panels diagnose model adequacy; they do not upgrade nominal marginal coverage to a complete uncertainty claim.**TABLE 5.** Median B2 hidden-voltage HVRE at the largest mismatch scale, $m = 1.5$. Ratios use the 140-case main-grid nominal median of 0.014744%.

Mismatch family	HVRE (%)	Ratio
M1 network	0.250430	16.985×
M2 machine	0.014744	1.000×
M3 governor	0.014743	1.000×
M4 AVR	0.014744	1.000×
M5 load model	0.014761	1.001×
M6 operating point	0.122641	8.318×
M7 coupled	0.023615	1.602×

TABLE 6. Sign-pooled weak load-event TEST results for the frozen W2/L2 model. Each row contains 1,600 event cases (16 source buses, two signs, 50 noise replicas); the false-positive rate uses 800 no-event records.

$ a $ (%)	AUROC	FPR (%)	FNR (%)	Top-1 (%)	Top-3 (%)
0.010	0.733	0.25	83.1	13.0	37.1
0.020	0.906	0.25	45.0	31.8	69.0
0.050	0.995	0.25	4.5	73.4	95.9
0.100	1.000	0.25	0.2	93.2	99.4
0.200	1.000	0.25	0.0	98.9	100.0
0.400	1.000	0.25	0.0	100.0	100.0

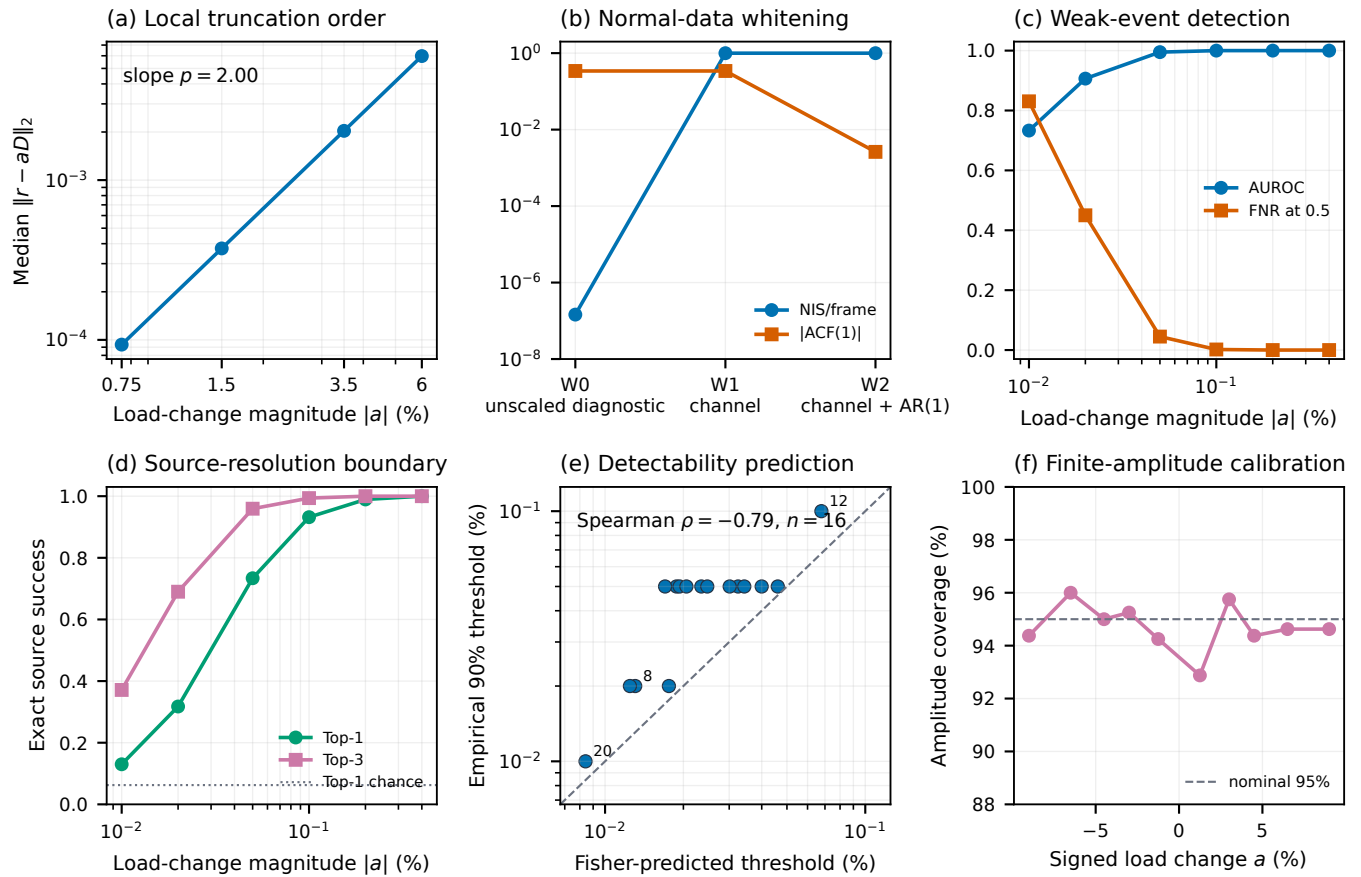


FIGURE 2. Single-source development evidence for the frozen load likelihood. The figure is retained as component validation; GLOBAL-137 in the main paper is the prospective support-recovery result.

TABLE 7. Audited sequential event-diagnosis baseline on the known-source TEST split, mean over three fitted seeds. The independent split unit is the complete trajectory; timestamp-level scores retain the denominators of the frozen evaluator.

Configuration	Detection F1	Event macro-F1	Physical Top-1	Physical Top-3	Integrity Top-1	False alarms/min
136-feature hierarchy	0.965	0.791	0.496	0.646	0.870	4.01
448-feature hierarchy	0.975	0.815	0.590	0.717	0.865	1.45
448 features + availability	0.989	0.905	0.592	0.719	0.974	1.45

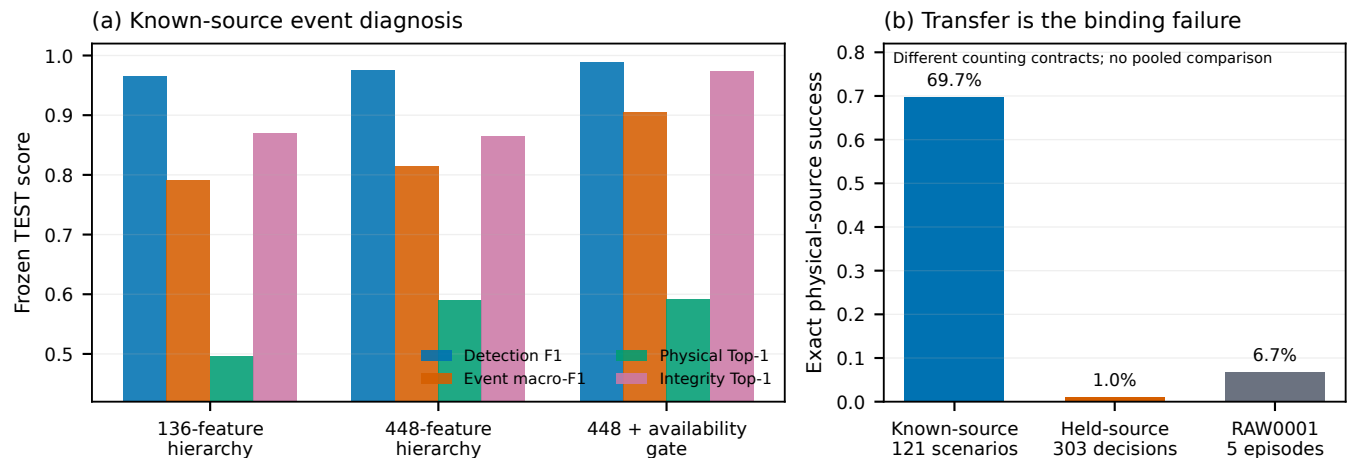


FIGURE 3. Secondary evidence for the supervised ANDES baseline, including known-source performance and the source-holdout failure.

REFERENCES

- [1] M. Katanic, J. Lygeros, and G. Hug, "Recursive dynamic state estimation for power systems with an incomplete nonlinear DAE model," *IET Generation, Transmission & Distribution*, vol. 18, no. 22, pp. 3657–3668, 2024.
- [2] S. A. Nugroho, A. F. Taha, N. Gatsis, and J. Zhao, "Observers for differential algebraic equation models of power networks: Jointly estimating dynamic and algebraic states," *IEEE Transactions on Control of Network Systems*, vol. 9, no. 3, pp. 1531–1543, 2022.
- [3] B. Qu, Z. Wang, B. Shen, H. Dong, and H. Liu, "Event-based joint state and unknown input estimation for energy networks: Handling multi-machine power grids," *IEEE Transactions on Network Science and Engineering*, vol. 10, no. 1, pp. 253–264, 2023.
- [4] Q.-H. Ngo, B. L. H. Nguyen, T. V. Vu, J. Zhang, and T. Ngo, "Physics-informed graphical neural network for power system state estimation," *Applied Energy*, vol. 358, p. 122602, 2024.
- [5] H. Yue, W. Zhang, U. C. Yilmaz, T. Yildiz, H. Huang, H. Liu, Y. Lin, and A. Abur, "Graph-learning-assisted state estimation using sparse heterogeneous measurements," *Electric Power Systems Research*, vol. 235, p. 110644, 2024.
- [6] S. Moshtagh, B. Azimian, M. Golgol, and A. Pal, "Topology-aware graph neural network-based state estimation for PMU-unobservable power systems," *IEEE Transactions on Power Systems*, 2025.
- [7] S. Pandey, A. K. Srivastava, and B. G. Amidan, "A real time event detection, classification and localization using synchrophasor data," *IEEE Transactions on Power Systems*, vol. 35, no. 6, pp. 4421–4431, 2020.
- [8] T. Yildiz and A. Abur, "Convolutional neural network-assisted fault detection and location using few PMUs," *Electric Power Systems Research*, vol. 235, p. 110705, 2024.
- [9] Z. Li, J. Tian, and H. Liu, "A joint data and knowledge-driven method for power system disturbance localisation," *IET Generation, Transmission & Distribution*, vol. 18, no. 24, pp. 4078–4089, 2024.
- [10] A. Kummerow, M. Dirbas, C. Monsalve, and P. Bretschneider, "Siamese sigmoid networks for the open classification of grid disturbances in power transmission systems," *IET Smart Grid*, vol. 6, no. 2, pp. 136–146, 2023.
- [11] Y. Hu and Z. Cheng, "Generative AI for open-set event detection and classification in power systems using synchrophasor data," *IEEE Transactions on Industrial Informatics*, 2026, early access.
- [12] M. Rimkus, P. Kokoszka, D. Duan, X. Wang, and H. Wang, "Graph neural networks for the localization of faults in a partially observed regional transmission system," *Scandinavian Journal of Statistics*, vol. 52, no. 2, pp. 572–594, 2025.
- [13] Z.-M. Zhai, M. Moradi, and Y.-C. Lai, "Detecting attacks and estimating states of power grids from partial observations with machine learning," *PRX Energy*, vol. 4, no. 1, p. 013003, 2025.
- [14] A. N. Montanari, L. Freitas, D. Proverbio, and J. Gonçalves, "Functional observability and subspace reconstruction in nonlinear systems," *Physical Review Research*, vol. 4, no. 4, p. 043195, 2022.
- [15] F. Pasqualetti, F. Dörfler, and F. Bullo, "Attack detection and identification in cyber-physical systems," *IEEE Transactions on Automatic Control*, vol. 58, no. 11, pp. 2715–2729, 2013.
- [16] M. Showkatbakhsh, Y. Shoukry, S. N. Diggavi, and P. Tabuada, "Securing state reconstruction under sensor and actuator attacks: Theory and design," *Automatica*, vol. 116, p. 108920, 2020.
- [17] H. E. Rauch, F. Tung, and C. T. Striebel, "Maximum likelihood estimates of linear dynamic systems," *AIAA Journal*, vol. 3, no. 8, pp. 1445–1450, 1965.

...